

Binary cross-entropy is usually used along with Sigmoid activation function:

$$L = -\hat{y} \log \hat{y} - (1 - \hat{y}) \log(1 - \hat{y})$$

$$\hat{y} = \sigma(z)$$

$$z = wX + b$$

$$\frac{dL}{d\hat{y}} = -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})}$$

$$\frac{d\hat{y}}{dz} = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$$

$$\frac{dz}{dw} = X, \text{ and } \frac{dz}{db} = 1$$

$$\frac{dL}{dw} = \frac{dL}{d\hat{y}} \times \frac{d\hat{y}}{dz} \times \frac{dz}{dw}$$

$$= \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right) \times \hat{y}(1-\hat{y}) \times X$$

$$= \{ -y(1-\hat{y}) + \hat{y}(1-y) \} \times X$$

$$= (-y + y\hat{y} + \hat{y} - y\hat{y}) \times X = (\hat{y} - y) \times X$$

$$\frac{dL}{db} = \frac{dL}{d\hat{y}} \times \frac{d\hat{y}}{dz} \times \frac{dz}{db}$$

$$= \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right) \times \hat{y}(1-\hat{y}) \times 1$$

$$= (\hat{y} - y)$$

Therefore, the update rules for weights and biases are-

$$w_{\text{new}} = w - \eta \frac{dL}{dw} = w - \eta X(\hat{y} - y)$$

$$b_{\text{new}} = b - \eta \frac{dL}{db} = b - \eta(\hat{y} - y)$$